

A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously reactivates neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is considered to support systems consolidation, the process by which the retrieval of memory content becomes progressively more dependent on neocortical rather than hippocampal circuits. Evidence indicates that reactivation events are not indiscriminate but preferentially target specific memory traces.

Systems consolidation has motivated substantial modeling efforts, and existing models offer complementary accounts of spontaneous reactivation, the preferential replay of specific memory traces, and hippocampo-neocortical redistribution of memory representations. Here, we ask whether these phenomena, and their underlying neural implementation, can be jointly recovered from a single normative objective.

We model the hippocampo-neocortical complex as two coupled associative memory networks implemented within the predictive coding (PC) framework. We then define our objective as minimizing the largest neocortical reconstruction deficit over the hippocampal memory manifold.

We show that, under suitable simplifying assumptions, gradient-based optimization of this objective determines the coupling between the two networks and their intrinsic dynamics. The architecture implement targeted memory reactivation and progressively redistribute memory representations through a closed-loop dialogue between the two systems. We also show that under this dynamics, full memory content transfer between the two systems is significantly faster than for random reactivation. Moreover, the derived coupling circuitry align with biological data on the specific pathway and interactions modalities between the prefrontal cortex (PCF) and hippocampal Cornu Ammonis 1 (CA1).

Author summary

1 Introduction

A substantial fraction of the brain’s metabolic budget is devoted to intrinsic activity [1,2]. Intrinsic activity can be defined as any activity which is not dedicated the processing of current sensory signal or the control of overt behaviour [1,2]. This intrinsic activity is especially evident during sleep and quiet wakefulness, when interaction with the environment is reduced **yet the brain sustains highly structured spontaneous dynamics (weird flow too unformal)** [3,4]. Moreover, The brain’s overall metabolic demand remains substantial throughout sleep: glucose utilization decreases of only one quarter during non-rapid-eye-movement (NREM) sleep but returns towards waking levels during rapid-eye-movement (REM) sleep [5,6]. **Furthermore (maybe change)**, sleep and sleep-like states are widely conserved across animal lineages, with states satisfying their defining behavioural criteria described in widely separated taxa, including insects and nematodes [7]. The coexistence of substantial metabolic demand, broad phylogenetic conservation, and the absence of immediate behavioural outcomes has motivated a long-standing effort to identify the functions supported by this activity.

Among the events taking place during the brain’s intrinsic activity, **one of the best characterised (too unformal)** is neural reactivation: the reinstanciation, during sleep or quiet rest, of activity patterns observed during waking experience. Observations in the rodent hippocampus showed that place cells which was co-active while an animal occupied particular locations during awake activity exhibited an

increased tendency to be co-active again during subsequent sleep [8]. Replay has since been reported in humans during post-task rest, including tasks with no spatial component [9, 10]. Subsequent works established that replay is not confined to the hippocampus: reactivation of patterns observed during behavioural tasks has also been shown in the ventral striatum, prefrontal cortex and primary visual cortex [11–13]. Hippocampal reactivation is commonly observed during sharp-wave ripples, a subset of these events being temporally coordinated with neocortical replay, during slow oscillations and thalamocortical spindles, allowing reactivation in the hippocampus and neocortex to be aligned within a common temporal scaffold [14, 15]. This coordination is thought to provide windows favourable to memory processing and is commonly described as a hippocampo-cortical dialogue (REF).

This dialogue is often considered as supporting system consolidation, the process by which the retrieval of memory during cognitive tasks becomes progressively less dependent on the hippocampus and more dependant on the neocortical circuits [?, 4, 14]. Moreover, this reorganization has been associated to the maturation of memory engram in the prefrontal cortex [?]. In this work, we frame system consolidation as a form of memory transfer, whereby memory content is partially or completely transfered from a subsystem to another.

Replay, and more broadly spontaneous activity during sleep, has been assigned many other functions, among which we have : Synaptic renormalisation and downscaling [?, ?], the extraction of statistical regularities and generalisation of learning across episodes [14], and metabolite clearance [?]. These views are not mutually exclusive and this work specifically address the memory transfer phenomenon.

2 Models

[Define the two predictive-coding associative memories, local recipient inference and plasticity, the recipient deficit, the worst-case consolidation objective, the three timescales, and the three selection protocols. Keep this section compact and equation-led; put full derivations in S1 Appendix.]

2.1 Two predictive-coding associative memory systems

[Introduce a fixed teacher or source T , interpreted as a hippocampal-like fast memory, and a plastic student or recipient S , interpreted as a neocortical-like slower memory. Both are covariance-learning predictive-coding associative memories based on Tang et al. The teacher is below the student. The PFC comparison concerns an effective top-down control interface, not the entire cortical storage system.]

$$M_T = I - W_T, \quad M_S = I - W_S, \quad \mathcal{U}_T = \ker M_T. \quad (1)$$

$$\mathbb{S}_T = \{x_T \in \mathcal{U}_T : \|x_T\| = 1\}. \quad (2)$$

[Explain that $\mathbb{S}_T$ is a mathematical comparison domain. Teacher recurrence and normalization approximate this hard constraint in simulations, and manifold leakage is measured.]

2.2 Recipient inference and local plasticity

With $\varepsilon_{TS} = x_S - x_T$ and $\varepsilon_S = M_S x_S$,

$$F_S(x_S, x_T; W_S) = \frac{\pi_{TS}}{2} \|\varepsilon_{TS}\|^2 + \frac{\pi_S}{2} \|\varepsilon_S\|^2, \quad (3)$$

$$\tau_S \dot{x}_S = -\pi_{TS} \varepsilon_{TS} - \pi_S M_S^\top \varepsilon_S, \quad \dot{W}_S = \eta \pi_S \varepsilon_S x_S^\top. \quad (4)$$

[State briefly that this is recipient-state descent and local recurrent-error plasticity. It is setup inherited from the predictive-coding associative-memory substrate, not the main result.]

2.3 Recipient deficit and worst-case consolidation

$$q(x_T; W_S) = \min_{x_S} F_S(x_S, x_T; W_S), \quad \mathcal{D}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S). \quad (5)$$

$$\min_{W_S} \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \quad (6)$$

[Motivate the maximum rather than an average: future memory demands and source availability are uncertain, and mean performance can hide a large recipient blind spot. The claim is that local dynamics implement the associated nested computation, not that the brain explicitly represents $\mathcal{D}_{\max}$.]

2.4 Timescales, selection controls, and rectification

$$\tau_S \ll \tau_T \ll 1/\eta : \quad \min_{x_S} \longrightarrow \max_{x_T \in \mathbb{S}_T} \longrightarrow \min_{W_S}. \quad (7)$$

[Define a consolidation bout: freeze recipient plasticity; select a teacher state; let the recipient settle; apply one identical small recipient update; repeat. The recipient is never clamped. Compare autonomous teacher dynamics, a greedy oracle that maximizes q , and random teacher-memory clamping. Use matched networks, memories, update magnitudes, settling criteria, stopping rules, and seeds.]

[Introduce a general activation f with $f(m_p) = m_p$. Identity gives the analyzed linear model. ReLU without bias with non-negative patterns gives the tested piecewise-linear extension. It can constrain reactivation toward individual memories but does not create ordered replay sequences.]

[figure placeholder]

Fig 1. Biological motivation and model architecture. Panel A: awake prefrontal control suppresses unwanted or competing hippocampal retrieval. Panel B: proposed reuse during NREM, where recipient competence suppresses already consolidated content. Panel C: the two associative memories, local errors, coupling κ , plasticity, and three timescales. Panel D: recipient deficit $\rightarrow$ teacher selection $\rightarrow$ recipient learning $\rightarrow$ self-extinction.

3 Results

[First derive the deficit object and coupling sign; then show autonomous transfer in a transparent three-memory example; then test priority, familiarity, efficiency, robustness, and rectification.]

3.1 The settled deficit identifies the least represented teacher-supported direction

$$S_S = M_S^\top M_S, \quad N_S = \pi_S S_S (\pi_{TS} I + \pi_S S_S)^{-1}. \quad (8)$$

Proposition 1 (Settled deficit and spectral certificate). *For every $x_T \in \mathbb{S}_T$, the linear recipient has unique settled state*

$$x_S^* = (I - N_S)x_T, \quad q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T. \quad (9)$$

If U_T is an orthonormal basis of $\mathcal{U}_T$ and $A_S = U_T^\top N_S U_T$, then

$$\mathcal{D}_{\max} = \frac{\pi_{TS}}{2} \lambda_{\max}(A_S). \quad (10)$$

[Explain that N_S measures unresolved content in the full state space and A_S restricts it to teacher-supported memories. The leading eigendirection is the largest current recipient deficit. Move scalar-mode and Rayleigh-Ritz proofs to S1 Appendix.]

3.2 Maximizing recipient deficit derives negative teacher-side coupling

$$\nabla_{x_T} q = \nabla_{x_T} F_S|_{x_S=x_S^*} = -\pi_{TS}\varepsilon_{TS}^*. \quad (11)$$

$$\tau_T \dot{x}_T|_{\text{interface}} = \kappa \varepsilon_{TS}, \quad \boxed{\kappa = -\mu_T \pi_{TS} < 0.} \quad (12)$$

[Keep this short derivation in the main paper. Teacher dynamics expose the recipient's deficit, while recipient inference and plasticity reduce it. Do not reintroduce the removed saddle, zero-sum representation, or global free-energy claim.]

Theorem 1 (Local implementation of worst-case consolidation). *Under hard teacher-memory constraint, separated recipient-inference, teacher-selection and recipient-plasticity timescales, and a nonzero initial component in the leading discrepancy eigenspace, projected teacher dynamics ascend the settled recipient deficit and approach its maximizing eigenspace. At a simple leading eigenvalue, the subsequent local recipient update descends $\mathcal{D}_{\max}$ and gives the steepest feasible first-order decrease among sufficiently small updates of equal norm.*

3.3 Learning removes the signal that selected a memory

Corollary 1 (Completion and self-extinction).

$$\mathcal{D}_{\max} = 0 \iff \mathcal{U}_T \subseteq \ker M_S. \quad (13)$$

At completion, $x_S^* = x_T$, $\varepsilon_{TS}^* = 0$, and $\varepsilon_S^* = 0$ for every valid teacher state. Both teacher selection and recipient plasticity vanish.

[Connect this to the biological claim: learning removes the discrepancy that generated priority, without a separate decay variable.]

3.4 Autonomous reactivation transfers a three-memory manifold

[Open with an elegant three-memory system in three dimensions. Explain the behavior before spectra, sweeps, or efficiency comparisons. In the linear condition, state explicitly that the network transfers a subspace and may express memory mixtures.]

[figure placeholder]

Fig 2. Autonomous transfer in a three-memory system. Panel A: teacher memories and manifold in three dimensions. Panel B: teacher and settled recipient trajectories across bouts. Panel C: recipient reconstruction or recurrent-weight heat map. Panel D: $\mathcal{D}_{\max}$, interface error, and recipient recurrent error until transfer self-extinguishes.

3.5 Recipient deficits organize priority and recipient-relative familiarity

[Show target switching: the teacher first occupies the largest deficit, then shifts after that direction is learned. Compare naive, partially familiar, and fully familiar recipients with the teacher fixed. State that priority falls with recipient competence conditional on source support; source reliability remains a distinct variable.]

[figure placeholder]

Fig 3. Priority, target switching, and familiarity. Panel A: teacher-restricted novelty eigenvalues. Panel B: direction-by-bout teacher occupancy. Panel C: plasticity and reactivation probability. Panel D: naive, partially familiar, and fully familiar recipients.

3.6 Differential waking learning prioritizes successfully encoded weak memories

[During waking, use fast hippocampal and slow cortical plasticity to learn one orthogonal memory extensively and a second only until its hippocampal reconstruction error is low while its cortical error remains high. With order counterbalanced, test whether subsequent offline dynamics preferentially reactivate the second, source-supported but cortically weak memory.]

[figure placeholder]

Fig 4. Waking learning strength and subsequent reactivation. Panel A: exposure protocol. Panel B: hippocampal and cortical reconstruction errors before sleep. Panel C: memory-specific reactivation occupancy during subsequent offline consolidation.

3.7 Autonomous prioritization approaches an oracle and outperforms random reactivation

[Primary efficiency test. Under the common discrete protocol, determine whether autonomous selection approaches the greedy oracle while reducing the worst-case deficit more efficiently than random teacher-memory selection.]

[figure placeholder]

Fig 5. Prioritized versus oracle versus random transfer. Panel A: $\mathcal{D}_{\max}$ versus identical plasticity events. Panel B: $\mathcal{D}_{\max}$ versus cumulative recipient-weight change. Panel C: novelty-spectrum evolution. Panel D: events to fixed thresholds and autonomous-to-oracle selection gap. Panel E: final transfer quality with matched-seed uncertainty.

3.8 Robustness and rectification

[Main text: summarize reliability across tested memory ranks, dimensions, correlations, partial familiarity, timescale separations, noise levels, and teacher-manifold leakage. Supporting Information: full sweeps. Use “reliable across tested regimes,” not a global convergence claim.]

[For ReLU without bias and non-negative stored patterns, show early trajectories near individual memories, later weakening as they are learned, time-resolved nearest-memory distance, cone confinement, and honest final-transfer quality.]

[figure placeholder]

Fig 6. Rectified dynamics and individual-memory reactivation. Panel A: linear signed-mixture geometry versus non-negative ReLU cone. Panel B: early and late trajectories. Panel C: time-resolved nearest-memory distance. Panel D: cone confinement. Panel E: transfer quality and self-extinction relative to the linear network.

4 Discussion

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; the negative effective interaction follows from maximizing recipient deficit; and learning makes prioritization self-terminating.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 State dynamics and diagnostics

[Teacher and recipient equations, activation, integration scheme, timestep, timescales, coupling, noise, normalization, settling, stopping, manifold leakage, and terminal occupancy.]

6.3 Prioritized, oracle, and random protocols

[Executable pseudocode or an algorithm box. Matching initializations, candidate states, one plasticity event per bout, thresholds, seed count, and uncertainty summaries.]

6.4 Outcome measures

[Define N_S , A_S , $\mathcal{D}_{\max}$, errors, occupancy, cumulative weight change, threshold events, oracle gap, nearest-memory distance, and cone confinement.]

Supporting information

- **S1 Appendix.** Full proofs of Proposition 1, Theorem 1, and Corollary 1; projected power iteration; envelope steps; antipodal maximizers; and feasible projected descent.
- **S1 Fig.** Positive, zero, and negative teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.

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